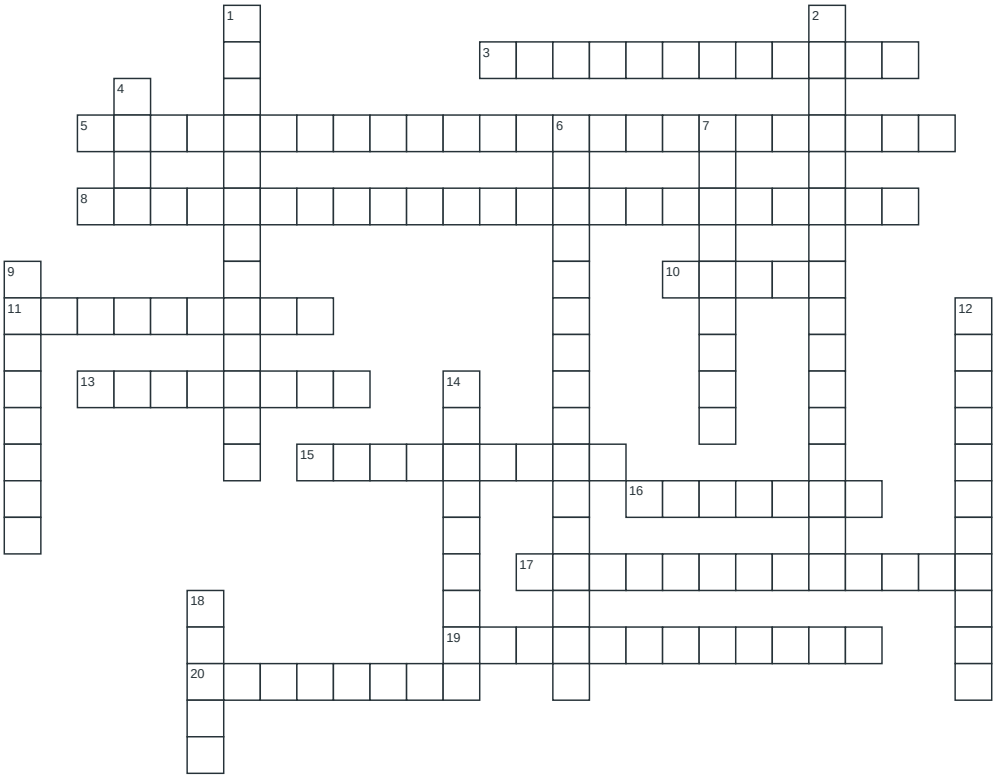
Wave Interference and Resonance Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

3. Pattern formed by overlapping waves traveling in opposite directions.
5. Overlapping displacements reinforce each other
8. Opposite displacements partly or completely cancel at overlap
10. Single, localized disturbance that travels through a medium.
11. The maximum displacement from equilibrium
13. Perception associated with sound intensity and hearing conditions.
15. Distance traveled by a wave per unit time.
16. Reduces oscillation amplitude over time
17. Increase in displacement when overlapping waves have the same phase.
19. Reduction in displacement when waves overlap with opposite phases.

ACROSS (continued)

20. Point of maximum displacement in a standing wave.

DOWN

1. Rate of loudness variation produced by two nearby sound frequencies.
2. The combination of overlapping wave displacements
4. A point with zero displacement amplitude
6. Frequency at which a system tends to oscillate when disturbed.
7. The number of cycles per unit time
9. Frequency that is an integer multiple of a system’s fundamental frequency.
12. Repeated motion or change around an equilibrium state.
14. A strong response when driving frequency is near a system's natural frequency
18. Position within a repeating wave cycle.

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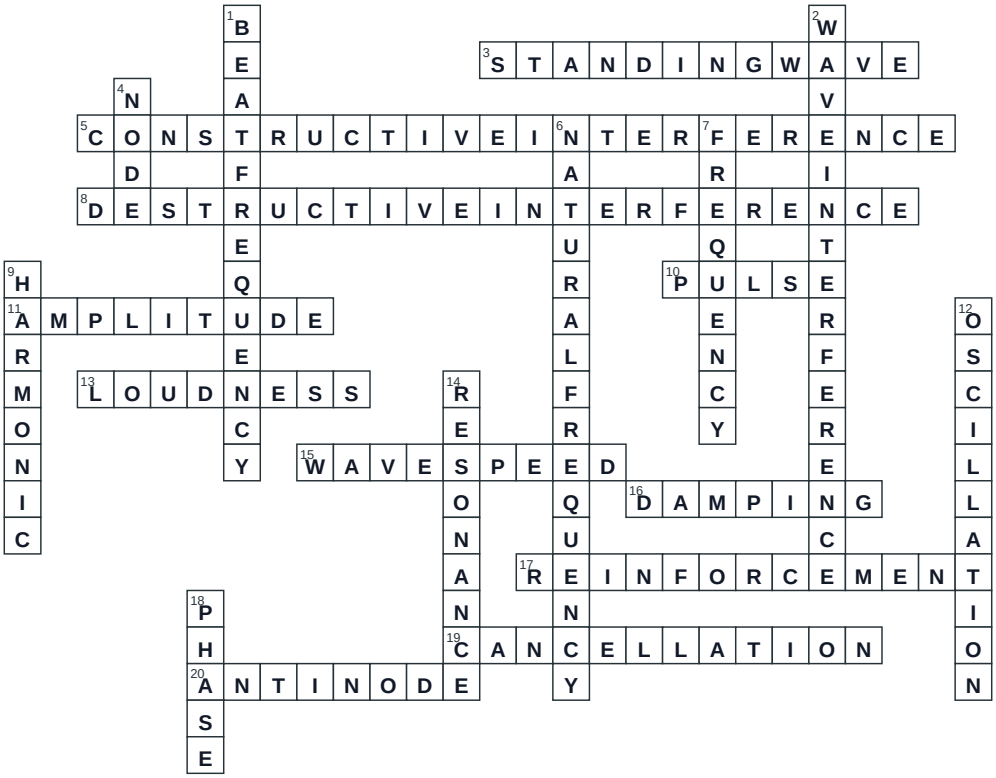
Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

3. Standing wave
5. Constructive Interference
8. Destructive Interference
10. Pulse
11. Amplitude
13. Loudness
15. Wave speed
16. Damping
17. Reinforcement
19. Cancellation
20. Antinode

DOWN

1. Beat frequency
2. Wave Interference
4. Node
6. Natural frequency
7. Frequency
9. Harmonic
12. Oscillation
14. Resonance
18. Phase